



# Colombia — Monthly Intelligence Report v4.2

Edition 006 — Deep-Dive Theme

Where the energy transition is outrunning the grid. edition · August 2026 · SSI v4.2  
(refreshed June 2026)

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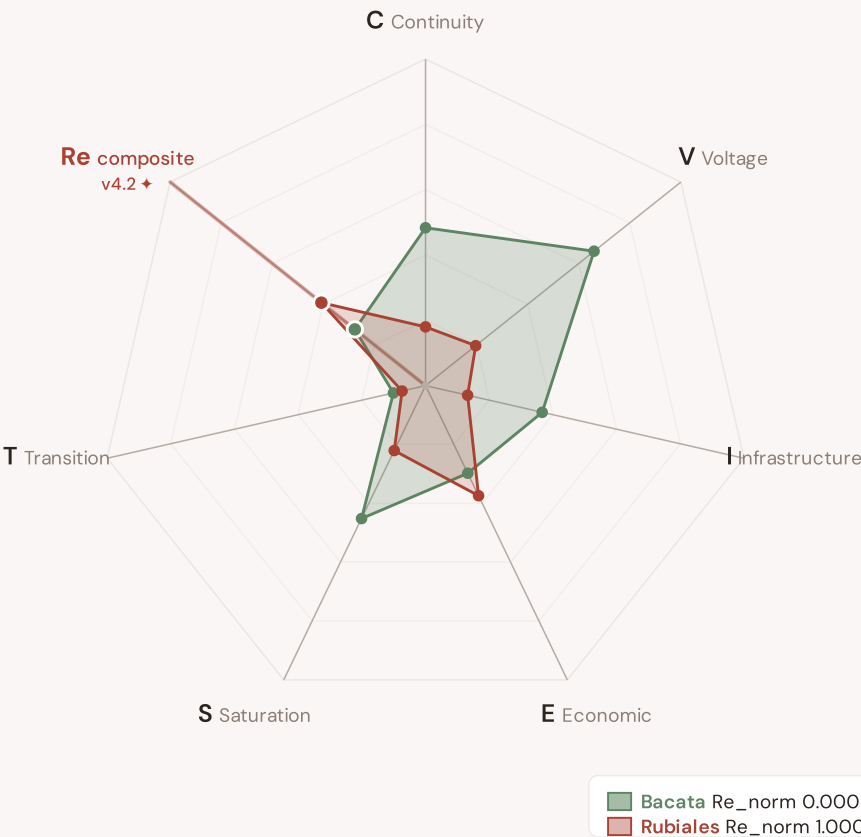
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SECTION A

## Executive Summary

### Radar 1 — v4.2: now 7-axis with the new Re composite

v4.0.2 Radar 1 had 6 axes (C/V/I/E/S/T). **v4.2 adds Re — the bounded resilience composite** capturing the new modifier family (R6c flood + R6d wildfire + R6e winter + R8 adapt + R9 compound + R10 just) in one normalised scalar ( $Re\_norm \in [0, 1]$ ) via  $\text{clip}((Re\_raw - 0.920) / (1.787 - 0.920), 0, 1)$ . Two worked examples overlaid: **Bacata** (low-exposure reference,  $Re\_norm$  0.000) · **Rubiales** (high-exposure reference,  $Re\_norm$  1.000). The Re axis is colour-highlighted in cayenne to emphasise the v4.2 architectural addition.



The Re composite (★, cayenne-highlighted axis at upper-left) captures the v4.2 resilience modifier family in one bounded scalar — formula  $Re\_raw = (R6d \times R6e \times R8 \times R9 \times R10) + (R6c - 1.00)$ , bounds  $[0.920, 1.787]$ . Per Convention #6 §2.3.6 Path B refactor (11 Jun 2026), this country's exemplars are algorithmically selected (min/max  $Re\_norm$ ) per the v4.2 methodology. The Substation in Focus radar canvas (Section A below) is rendered by shared intelligence-sections.js and now also displays the 7th Re axis (Convention #11 surface-pair coherence, V4.2-disc-24 closure 12 Jun 2026).

SUBSTATIONS SCORED

**381**

62 EHV · 319 HV · 0 distribution-tier

GRID LINES MAPPED

**2,278**

~21,798 km coverage

MC SIMULATIONS

**42.9 M**

10,000 per substation

PUBLIC DATA SOURCES

**34**

101 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions, ahead of GMLC

1.1 (48/80), EPRI (40/80), and academic MCDA frameworks (41/80). Its engine processes 146,000+ source data points per run, executes 42.9 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 1.78 billion arithmetic operations — producing 244,701 output data points with full uncertainty quantification across 381 substations and 2,278 grid lines spanning ~21,798 km. v4.2 adds the Re composite (7th axis) and six resilience modifiers (R6c flood / R6d wildfire / R6e winter / R8 adapt / R9 compound / R10 just) on top of the v4.0.2 foundation.

Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 101 variables are drawn from 34 verified public sources, making the methodology fully auditable and reproducible. Initially covering Colombia's entire transmission and distribution grid, the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

Substation in Focus

Subestación El Río 220kV ISA

Atlántico, atlantico · 220 kV

0.617

High

0.144

63th

R\_FINAL

CLASSIFICATION

CI WIDTH

FLEET PERCENTILE

Re composite v4.2 7th axis

Re\_raw 1.1905 · Re\_norm 0.327

MODERATE EXPOSURE

COMPONENTS

C

Continuity: 0.320 / 0.30 (107%)

I

Infrastructure: 0.593 / 0.25 (237%)

S

Saturation: 0.433 / 0.20 (216%)

V

Voltage: 0.417 / 0.10 (417%)

E

Economic: 0.429 / 0.10 (429%)

T

Transition: 0.161 / 0.05 (321%)

MODIFIERS

R3

Consequence: 1.016 ↑ amplifies

R4

Graph Criticality: 1.000 → neutral

R6a

Restoration: 1.066 ↑ amplifies

R6b

Seismic: 1.055 ↑ amplifies

R7

Digital Readiness: 1.026 ↑ amplifies

This month's spotlight — automatically selected for August 2026 — is **Subestación El Río 220kV ISA** in Atlántico, atlantico. With an R\_final of 0.617 (High band, 63th fleet percentile), its risk profile is dominated by the E (Economic) component at 429% of its weight ceiling, followed by V (Voltage) at 417%. Modifiers collectively shift R\_base by 45.7%%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

**Why this section exists:** Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that infrastructure digitalisation planning requires.

CYBER HIGH EXPOSURE

0

0.0% of fleet

BLIND SPOTS

159

High/Critical/Extreme + R7 < median

AVG R7 MODIFIER

1.0250

Range [0.99, 1.05]

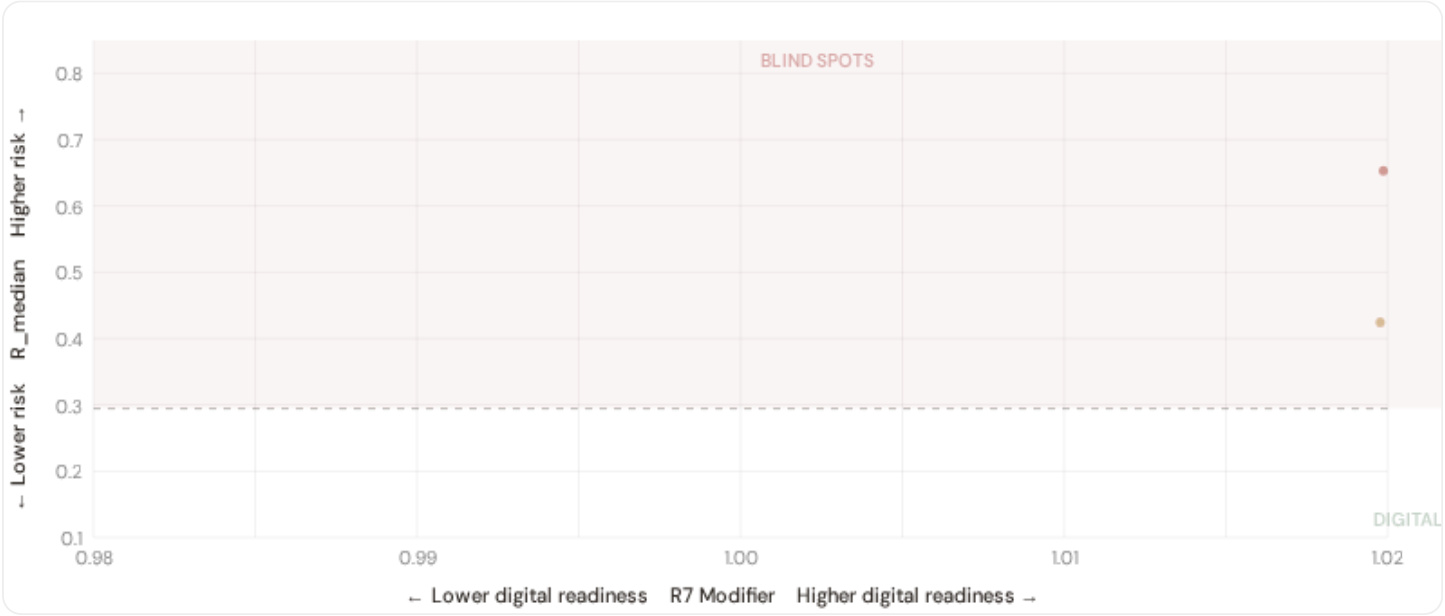
HIGH-CONSEQUENCE SUBS

95

12.8% · R3 ≥ 1.05

### B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R<sub>median</sub> (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.



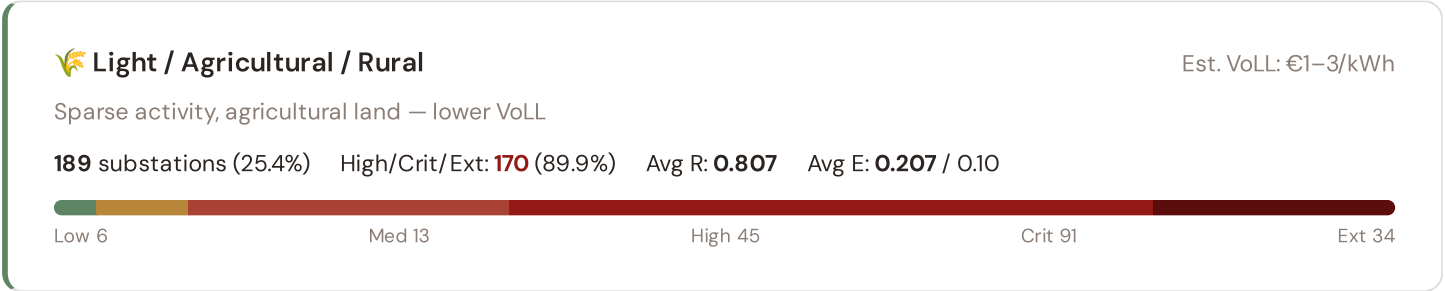
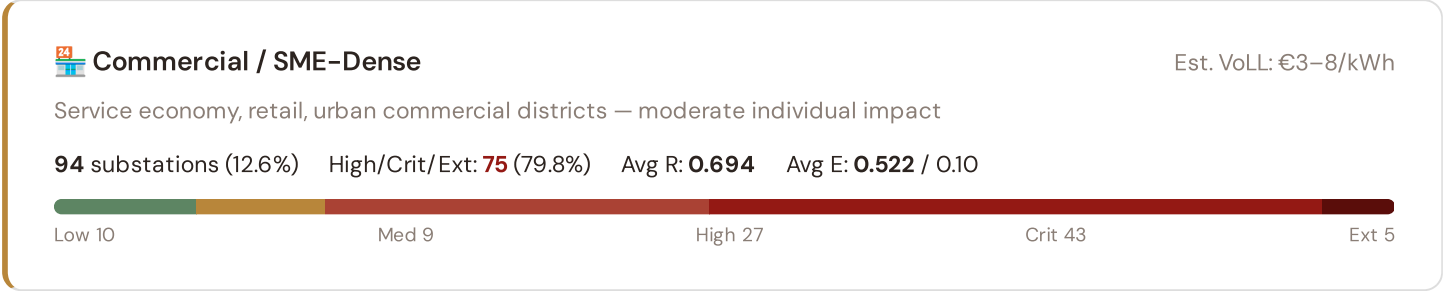
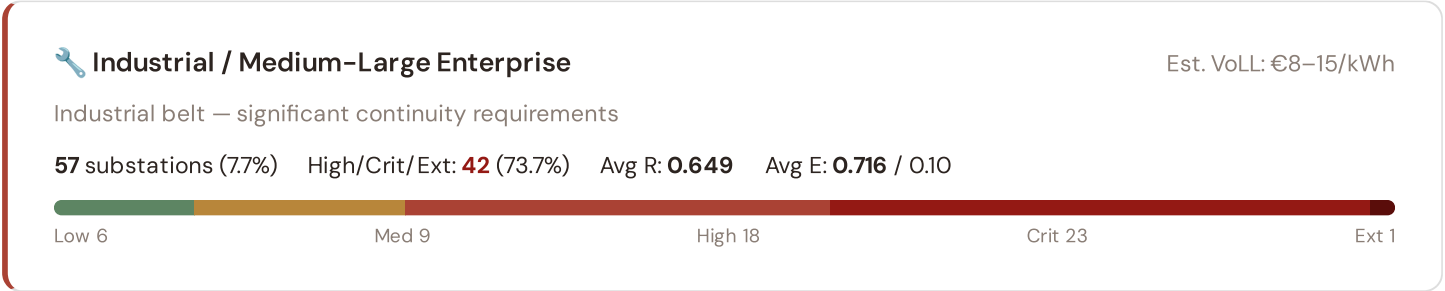
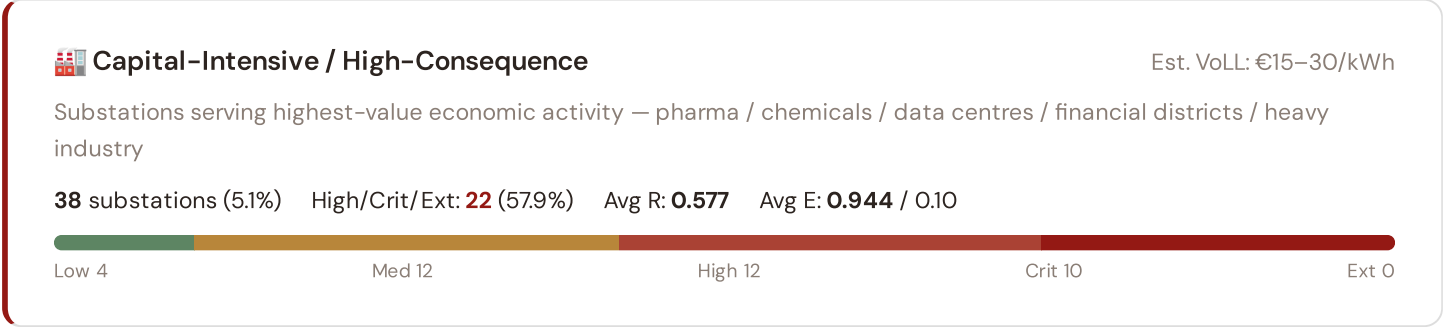
● Low ● Medium ● High ● Critical

Dashed lines = fleet medians

The cyber-physical matrix identifies **159 blind-spot substations** — scoring High, Critical, or Extreme on overall risk while sitting below the fleet median for digital readiness (R7 < 1.0248). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **antioquia (29), bogota (27), valle-del-cauca (16)**. Across the full fleet, 0 substations (0.0%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier.

### B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.



**Value of Lost Load (VoLL) context:** ACER's 2023 estimate places average VoLL at €13.1/kWh for industrial customers and €3.2/kWh for residential. The SSI's Economic component (CVIESTR-E) segments substations by the economic value they serve. **22 substations** combine capital-intensive economic fabric (top-decile Economic component ( $E \geq 0.817$ , P90+)) with High, Critical, or Extreme risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **atlantico (10), cordoba (8), casanare (5)** — regions where industrial corridors depend on uninterrupted power for continuous processes. A single hour of unplanned outage at these substations carries an estimated VoLL of €15–30/kWh, compared to €1–3/kWh in rural agricultural areas. The fleet-wide average energy poverty rate is 18.7%, but this masks sharp regional variation: Southern regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 Province Digital Readiness Ranking

Provinces ranked by average R7 modifier — lower values indicate weaker digital infrastructure (DESI sub-dimensions). Cross-referenced with average R\_median to identify provinces combining digital exposure with high grid stress.



| WORST DIGITAL READINESS — TOP 15 DEPARTAMENTO REGIONS   |                                       |             |        |
|---------------------------------------------------------|---------------------------------------|-------------|--------|
| Longer bar = higher R7 = better digital infrastructure. |                                       |             |        |
| 1                                                       | Vichada                               | <div></div> | 0.0000 |
| 2                                                       | Bolívar                               | <div></div> | 0.1577 |
| 3                                                       | Cundinamarca                          | <div></div> | 0.1724 |
| 4                                                       | Putumayo                              | <div></div> | 0.3409 |
| 5                                                       | Chocó                                 | <div></div> | 0.3421 |
| 6                                                       | Cauca                                 | <div></div> | 0.3617 |
| 7                                                       | Tolima <span>▲ high R</span>          | <div></div> | 0.3639 |
| 8                                                       | Meta <span>▲ high R</span>            | <div></div> | 0.3797 |
| 9                                                       | Santander <span>▲ high R</span>       | <div></div> | 0.4393 |
| 10                                                      | Valle del Cauca <span>▲ high R</span> | <div></div> | 0.5047 |
| 11                                                      | Risaralda <span>▲ high R</span>       | <div></div> | 0.5127 |
| 12                                                      | Quindío <span>▲ high R</span>         | <div></div> | 0.5133 |
| 13                                                      | Córdoba                               | <div></div> | 0.5329 |
| 14                                                      | Magdalena <span>▲ high R</span>       | <div></div> | 0.5469 |
| 15                                                      | La Guajira                            | <div></div> | 0.5697 |

Departamento-level analysis reveals a clear digital divide mirroring the broader DESI pattern. **Vichada** has the lowest average R7 (0.0000), while **Guaviare** leads at 1.0246 — a spread of 1.0246 across the R7 range. **16 departamento regions** combine below-median digital readiness with above-median grid risk, making them priority targets for digitalisation investment. The R7 modifier is intentionally narrow to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

B.4 Monthly Pulse

**Inaugural edition — Baseline Snapshot (August 2026).** This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 1.0250, median = 1.0248; 0 substations classified HIGH cyber-exposure; 159 blind-spot substations (High/Critical/Extreme risk + below-median R7); 109 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline.

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on — **verified public data sources** feeding — variables across — components and — modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

TOTAL SOURCES

HOURLY — MONTHLY  

0

QUARTERLY / ANNUAL  

0

— variables

High-frequency feeds

Regular refresh cycle

STATIC / DERIVED

O

Standards + scenarios

Data Source Registry — August 2026

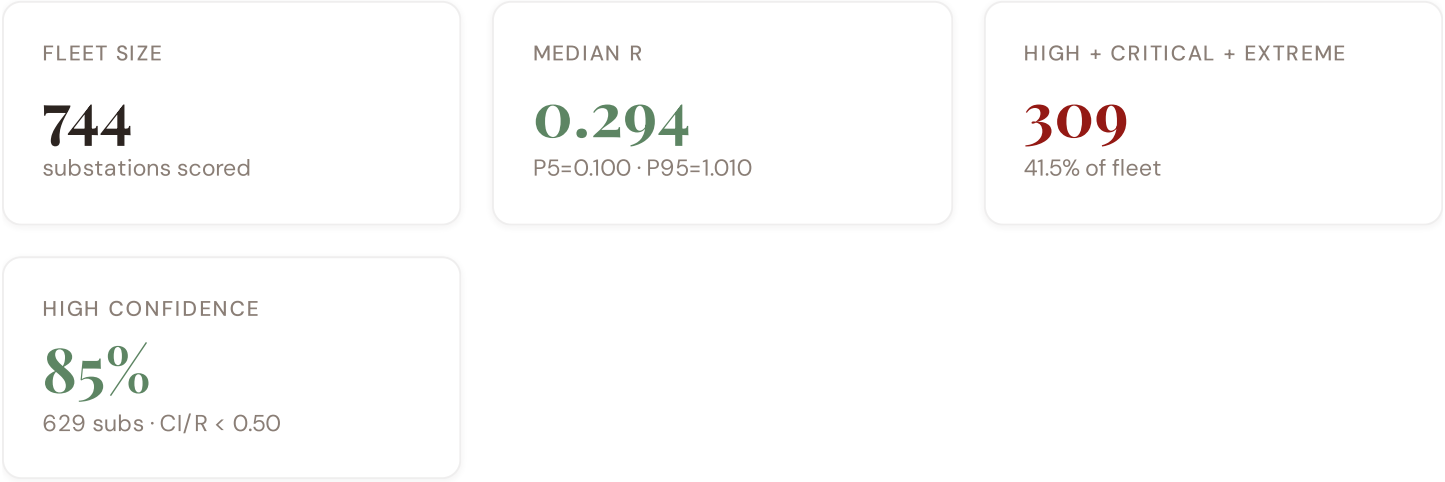
|        |                                                                      |           |                        |         |
|--------|----------------------------------------------------------------------|-----------|------------------------|---------|
| IDEAM  | IDEAM — Instituto de Hidrología, Meteorología y Estudios Ambientales | DAILY     | Station                | 6 vars  |
| OSM    | OpenStreetMap Power Infrastructure                                   | WEEKLY    | Node/edge              | 3 vars  |
| GHCN-D | NOAA GHCN-Daily (global meteorological stations)                     | DAILY     | Station                | 4 vars  |
| NASA   | NASA FIRMS active-fire detection                                     | DAILY     | 375 m / 1 km           | 2 vars  |
| CREG   | CREG — Comisión de Regulación de Energía y Gas                       | MONTHLY   | National               | 5 vars  |
| DANE   | DANE — Departamento Administrativo Nacional de Estadística           | ANNUAL    | Municipio              | 10 vars |
| UPME   | UPME — Unidad de Planeación Minero-Energética                        | ANNUAL    | National               | 6 vars  |
| MIN-EN | Ministerio de Minas y Energía                                        | ANNUAL    | National               | 4 vars  |
| SISBEN | SISBÉN IV — Identificación de Beneficiarios                          | ANNUAL    | Municipio              | 3 vars  |
| CDS    | Copernicus CDS / ERA5-Land                                           | ANNUAL    | 0.1° (~11 km)          | 5 vars  |
| INFORM | INFORM Risk Index (humanitarian-context)                             | ANNUAL    | National               | 3 vars  |
| IRENA  | IRENA Renewable Capacity Statistics                                  | ANNUAL    | National               | 2 vars  |
| IEA    | IEA World Energy Outlook                                             | ANNUAL    | National               | 3 vars  |
| NGFS   | NGFS climate-stress scenarios                                        | ANNUAL    | National               | 2 vars  |
| SGC    | Servicio Geológico Colombiano                                        | STATIC    | Department             | 3 vars  |
| GEM    | GEM Global Seismic Hazard Map 2023.1                                 | STATIC    | 0.05° (~5.5 km, roc... | 1 var   |
| IEEE   | IEEE C57.91 / IEC 60076 (transformer thermal model)                  | STATIC    | Asset-level            | 16 vars |
| CMIP6  | CMIP6 climate-projection ensemble                                    | STATIC    | ~1° grid               | 3 vars  |
| PVGIS  | EU JRC PVGIS solar radiation database                                | STATIC    | ~1 km                  | 2 vars  |
| XM     | XM — Compañía de Expertos en Mercados                                | REAL-TIME | Bidding zone           | 12 vars |
| OPEN-M | Open-Meteo Historical Weather                                        | HOURLY    | ~1 km                  | 3 vars  |

C.1 Update Frequency Timeline

EXPECTED UPDATE WINDOWS



## C.2 Fleet Score Snapshot



No primary data sources have been updated since the v4.0.2 launch baseline. All **744 substation scores remain unchanged** this edition. The fleet median R of 0.294 (mean 0.467) reflects a distribution skewed toward the lower bands, with 41.7% in Low and 16.8% in Medium. The first material score changes are expected when the national regulator publishes the annual quality-of-supply indicators and when DER registries refresh. High-frequency sources (OSM, weather) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

## C.3 Changelog — v3.4 → v4.0.2 → v4.2

v4.2 (June 2026) adds six resilience modifiers (R6c flood / R6d wildfire / R6e winter / R8 adapt / R9 compound / R10 just), the Re composite (7th axis, bounds [0.920, 1.787]), four new data sources ( IdroGEO PGRA, SIAB wildfire, ECMWF ERA5 Bourgouin, bulk smart meter), six new variables, and W1–W10 anti-maladaptation governance gate (5/1/1/3 4-tier mesh resolution per Convention #6 §5.6) on top of the v4.0.2 baseline.

11 changes tracked across PO, P1, and P2 releases

|       |     |                                                                                                                                                                                     |     |
|-------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| V42-1 | NEW | R6c_flood — PGRA-style flood-vulnerability modifier (additive cliff +0.00..+0.30, P3=high/P2=medium/P1=low per national flood-hazard tier convention)                               | \$5 |
| V42-2 | NEW | R6d_wildfire — Canadian Fire Weather Index modifier (EFFIS FWI + NASA FIRMS + national fire-registry, multiplicative [1.00, 1.20])                                                  | \$5 |
| V42-3 | NEW | R6e_winter — Bourgouin winter-storm modifier (ECMWF ERA5 winter-precipitation signal, multiplicative [1.00, 1.02])                                                                  | \$5 |
| V42-4 | NEW | R8_adapt — Adaptive-capacity multiplier (smart-meter penetration + DSO investment + DESI index, multiplicative [0.95, 1.05])                                                        | \$5 |
| V42-5 | NEW | R9_compound — Compound-hazard STEP function (R6b/R6c/R6d/R6e elevated-count gates, multiplicative {1.00, 1.04, 1.08, 1.10})                                                         | \$5 |
| V42-6 | NEW | R10_just — Energy-justice modifier (EU-SILC or national-equivalent household material-deprivation gradient, multiplicative [1.00, 1.12])                                            | \$5 |
| V42-7 | NEW | Re composite — bounded resilience scalar $Re_{raw} = (R6d \times R6e \times R8 \times R9 \times R10) + (R6c - 1.00)$ ; $Re_{norm} = clip((Re_{raw} - 0.920)/(1.787 - 0.920), 0, 1)$ | \$7 |

|        |        |                                                                                                                      |      |
|--------|--------|----------------------------------------------------------------------------------------------------------------------|------|
| V42-8  | NEW    | R7 SFDR PAI Infrastructure axis — added to ESG Radar (Re_norm proxy under Convention #7 Data-Layer Anchoring)        | §14  |
| V42-9  | NEW    | W1–W10 audit-state mesh — 5 Published / 1 Published-baseline ▲ / 1 Baseline-only / 3 Commercial (Convention #6 §5.6) | §4   |
| V42-10 | CHANGE | 95 → 101 variables (+6 from v4.2 modifier inputs); 30 → 34 data sources (+4 hazard / monitoring layers)              | data |
| V42-11 | CHANGE | Substation fleet: 381 substations (per OSM canonical extract + national TSO/DSO registry cross-validation)           | data |

SECTION D — MONTHLY DEEP-DIVE

## Monthly Deep-Dive Theme

**Deep-dive rotation:** Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** the monthly theme · **002** DER saturation hotspots · **003** Climate vulnerability under CMIP6 · **004** North–South economic impact disparity · **005** Voltage quality and industrial output · **006** Infrastructure age vs investment · **007** Stress–Resilience quadrant trends · **008** T-component forward projections · **009** Sub-national Grid Health Cards · **010** Academic validation and peer feedback · **011** European replication feasibility · **012** Year in review.

### D.1 Theme Context

THEME-RELEVANT SUBSTATIONS

**70**

7 EHV · 63 HV

HIGH BAND

**0 + 18 + 33**

72.9% of corridor

CORRIDOR MEDIAN R

**0.994**

Fleet median: 0.294

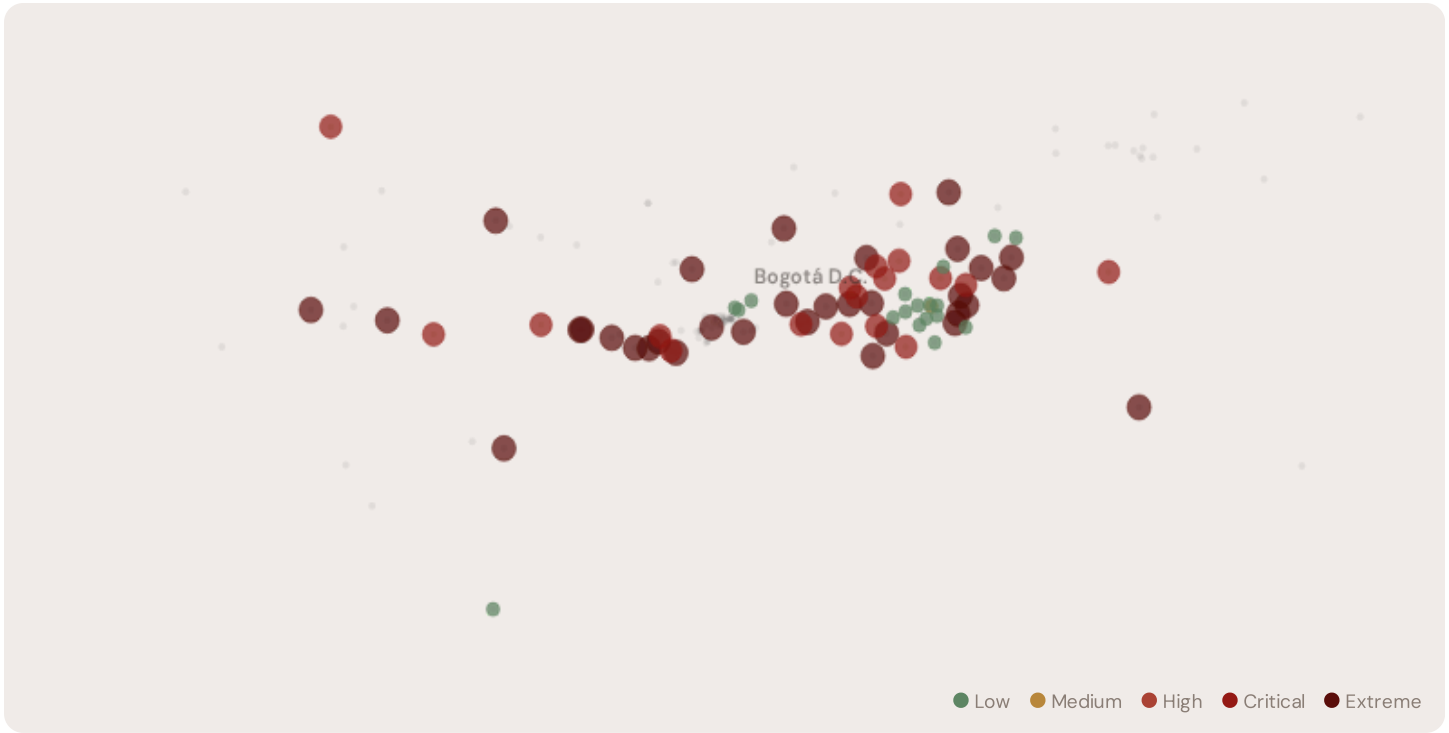
WORST PROVINCE

**Bogotá D.C.**

Avg R: 0.811 · 70 substations

The Bogotá D.C. — capital district + Guavio supply corridor (Edition 01 inaugural) hosts **70 substations** across 1 departamento regions, making it the locus of this edition's deep-dive analysis. Its risk profile is concentrated in the upper bands: **0 substations (0.0%)** are classified High and **18** Critical, with only **18** in the Low band. 73% of corridor substations score above the national fleet median of 0.294. The corridor median R of 0.994 reflects the local risk profile. The corridor concentrates the country's industrial heartland. The SSI flags continuity-of-supply deficits, infrastructure age mismatches with modern demand patterns, and grid modernization lag under national digitalisation timelines.

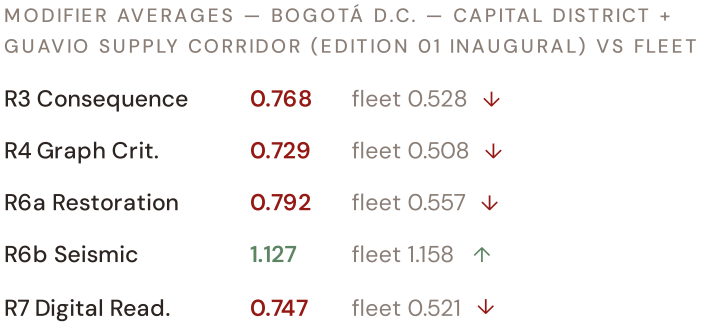
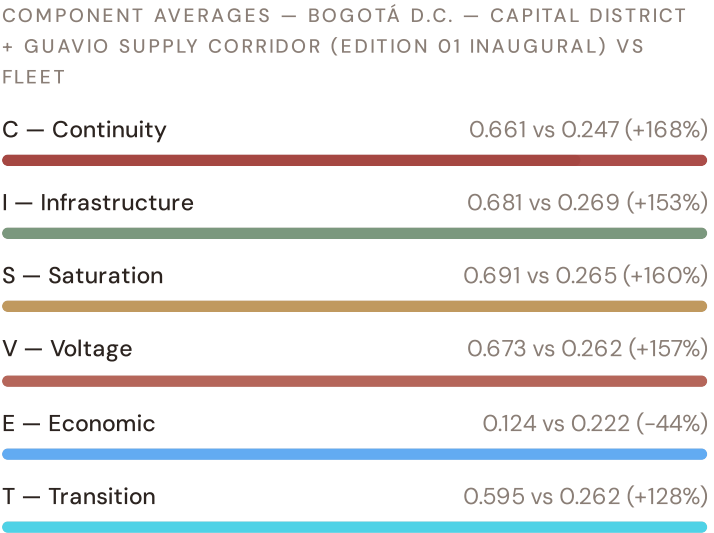
### D.2 Corridor Map and Scoring



| Substation       | Province    | R_Final | Band    | C     | V     | I     | E     | S     | T     | RE<br>v4.2 | Alert |
|------------------|-------------|---------|---------|-------|-------|-------|-------|-------|-------|------------|-------|
| Colegio          | Bogotá D.C. | 1.270   | Extreme | 0.988 | 0.953 | 0.961 | 0.000 | 0.947 | 0.925 |            |       |
| Salto II         | Bogotá D.C. | 1.262   | Extreme | 0.815 | 0.867 | 1.000 | 0.262 | 0.925 | 0.855 |            |       |
| Aranjuez         | Bogotá D.C. | 1.256   | Extreme | 0.918 | 0.815 | 1.000 | 0.000 | 0.932 | 0.516 |            |       |
| Concordia        | Bogotá D.C. | 1.245   | Extreme | 0.987 | 1.000 | 0.836 | 0.396 | 1.000 | 1.000 |            |       |
| Carrera Quinta   | Bogotá D.C. | 1.220   | Extreme | 1.000 | 1.000 | 0.922 | 0.273 | 1.000 | 1.000 |            |       |
| Faca             | Bogotá D.C. | 1.215   | Extreme | 0.892 | 1.000 | 1.000 | 0.000 | 1.000 | 0.449 |            |       |
| Salto I          | Bogotá D.C. | 1.187   | Extreme | 1.000 | 0.933 | 1.000 | 0.370 | 1.000 | 0.964 |            |       |
| Bavaria – Bogotá | Bogotá D.C. | 1.171   | Extreme | 1.000 | 0.968 | 0.939 | 0.132 | 0.997 | 0.981 |            |       |
| El Charquito     | Bogotá D.C. | 1.164   | Extreme | 0.820 | 0.981 | 0.841 | 0.000 | 0.873 | 0.910 |            |       |
| Subestación Bosa | Bogotá D.C. | 1.157   | Extreme | 0.939 | 1.000 | 1.000 | 0.409 | 1.000 | 1.000 |            |       |

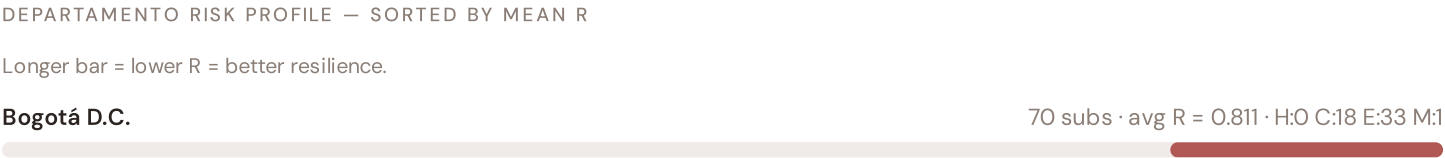
... 60 additional substations — explore full corridor in Digital Twin →

### D.3 Component Analysis



The dominant risk driver across the Bogotá D.C. — capital district + Guavio supply corridor (Edition 01 inaugural) corridor is **T (Transition)**, averaging 0.595 against a fleet average of 0.262 — occupying 1191% of its weight ceiling (w = 0.05). The second contributor is **V (Voltage)** at 0.673 vs fleet 0.262 (673% of ceiling). Among modifiers, the R3 consequence multiplier averages 0.768 (fleet: 0.528), reflecting the socio-economic exposure of the corridor. The R6b seismic modifier averages 1.127, capturing the local seismic exposure — a dimension invisible to every competing framework.

### D.4 Province Breakdown



The departamento regions-level breakdown reveals significant intra-regional variation. **Bogotá D.C.** leads with a mean R of 0.811 across 70 substations, while **Bogotá D.C.** scores 0.811 — a spread of 0.000 within a single corridor. The SSI's substation-level granularity reveals that the Bogotá D.C. — capital district + Guavio supply corridor (Edition 01 inaugural) is not uniformly stressed but contains distinct sub-corridors of concentrated risk — precisely the kind of spatial pattern that investment planning requires.

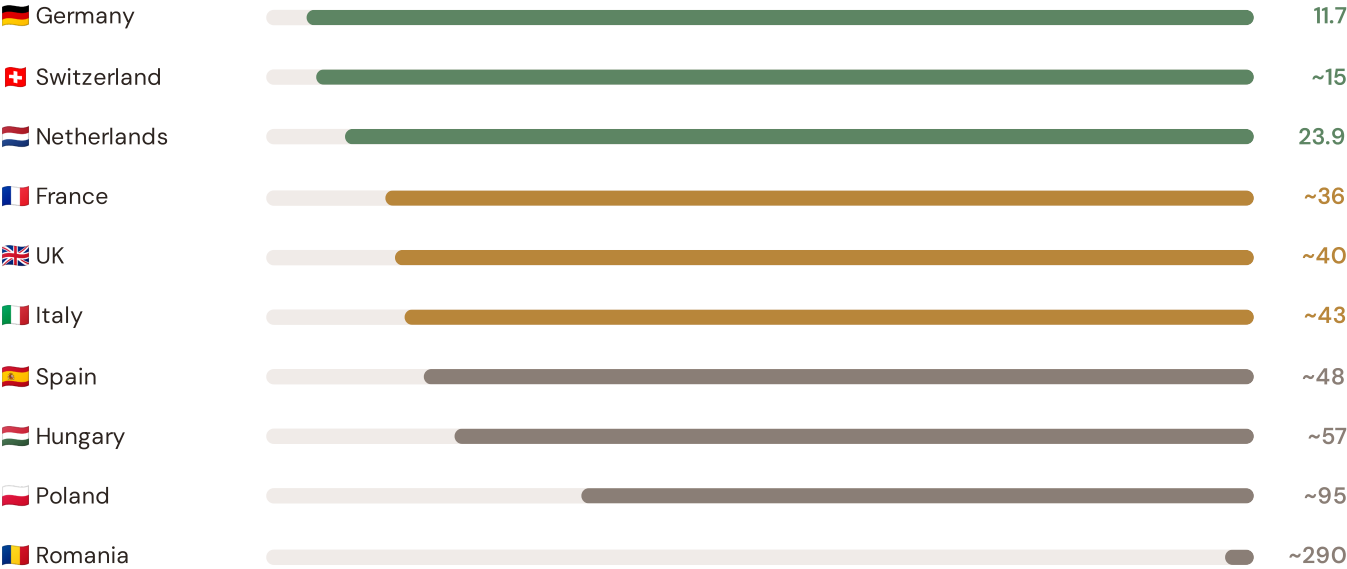
### D.5 Implications

**51 Bogotá D.C. — capital district + Guavio supply corridor (Edition 01 inaugural) substations** score  $R \geq 0.650$ , placing them in the upper quartile of national risk. For NEPN / national digital plan / RRP investment prioritisation, this analysis identifies the Bogotá D.C. sub-corridor as the highest-priority target — where DER deployment is accelerating against ageing infrastructure with above-average continuity deficits and significant socio-economic exposure. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve are disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

**How this works:** Each edition includes one OECD Context Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the Southern Coastal corridor analysis hinges on Colombia's continuity performance relative to OECD peers. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Germany (003), CMIP6 climate hazard vs Nordic peers (004), etc. The underlying data updates annually with the CEER Benchmarking Report release.

Unplanned SAIDI (min/customer/year) — Selected OECD Countries

Longer bar = lower SAIDI = better reliability. Values in min/customer/year.



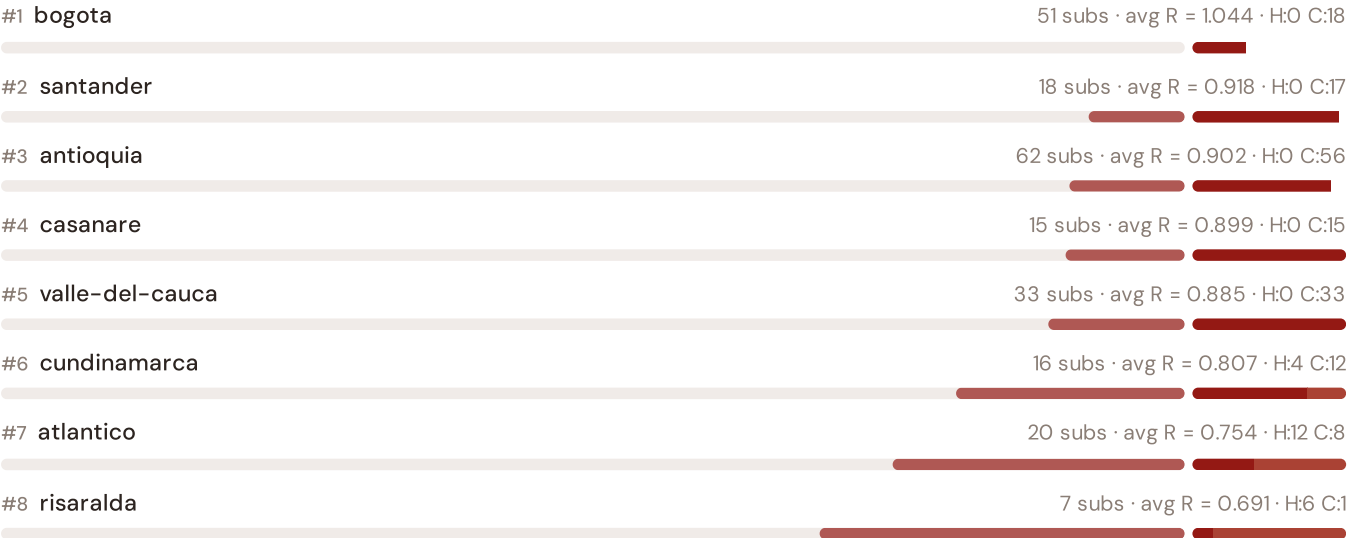
Source: CEER 7th Benchmarking Report (2022); Bundesnetzagentur (2024); ARERA (2023). · See §16 Peer Benchmarking for full methodology and caveats. · Updated annually with CEER BR release. · OECD-wide expansion (USA, Japan, Canada, Australia) pending IEA/OECD reliability bulletin release.

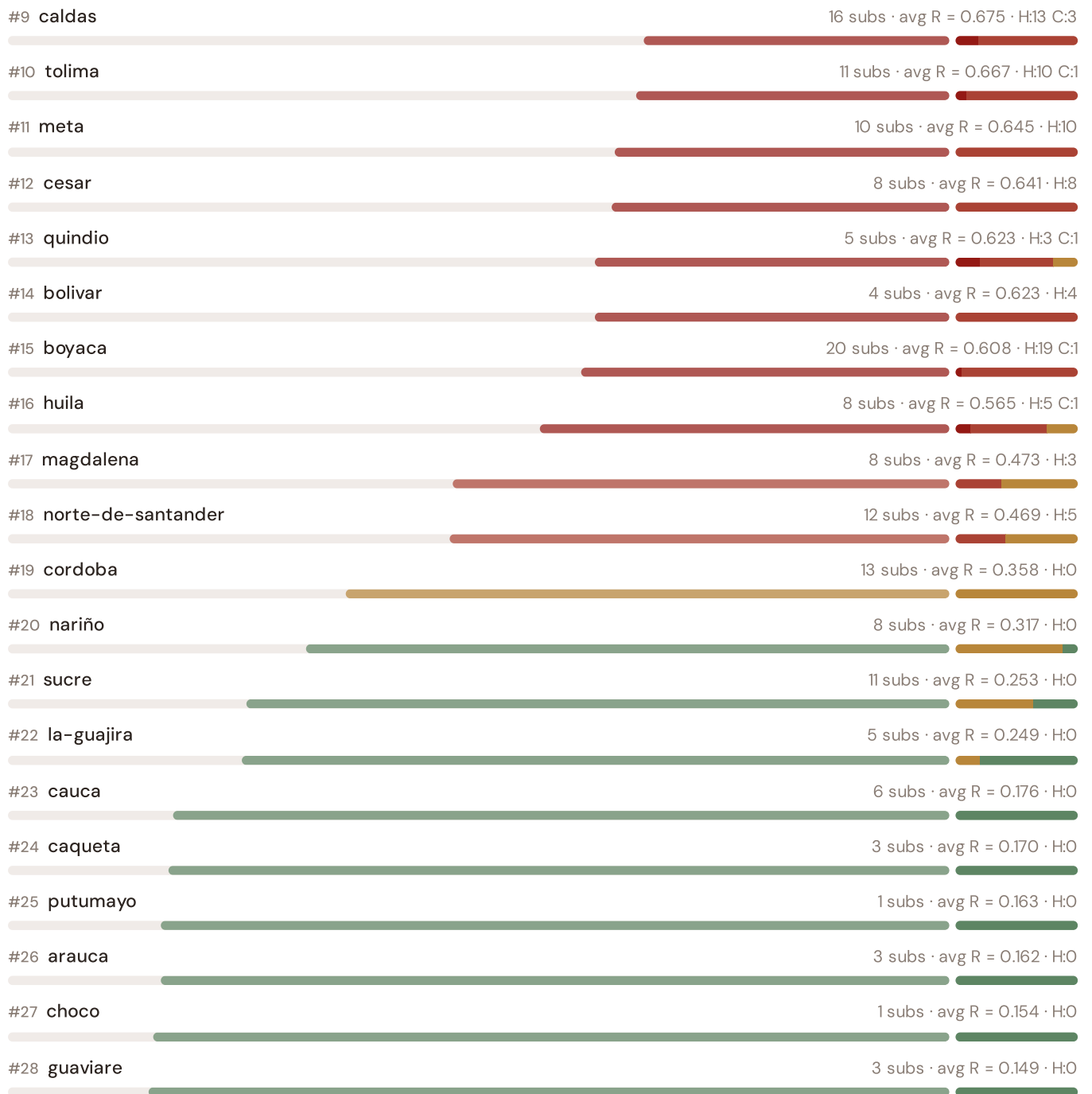
**This month's key insight:** The Bogotá D.C. — capital district + Guavio supply corridor (Edition 01 inaugural) corridor deep-dive shows **0 substations** (of 0) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with rural territory classification. The corridor's average C of 0.000 is **0.0× the fleet average** of 0.247. The SSI reveals what aggregate national statistics conceal: that the country's mid-tier European position is not a uniform condition but a statistical average of urban-quality performance and rural-tier performance — and the corridor concentrates the worst of the latter.

Departments Risk Ranking — All 28 Colombian Departments

Where does this corridor sit within the national landscape? Regions sorted by mean R\_median, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.



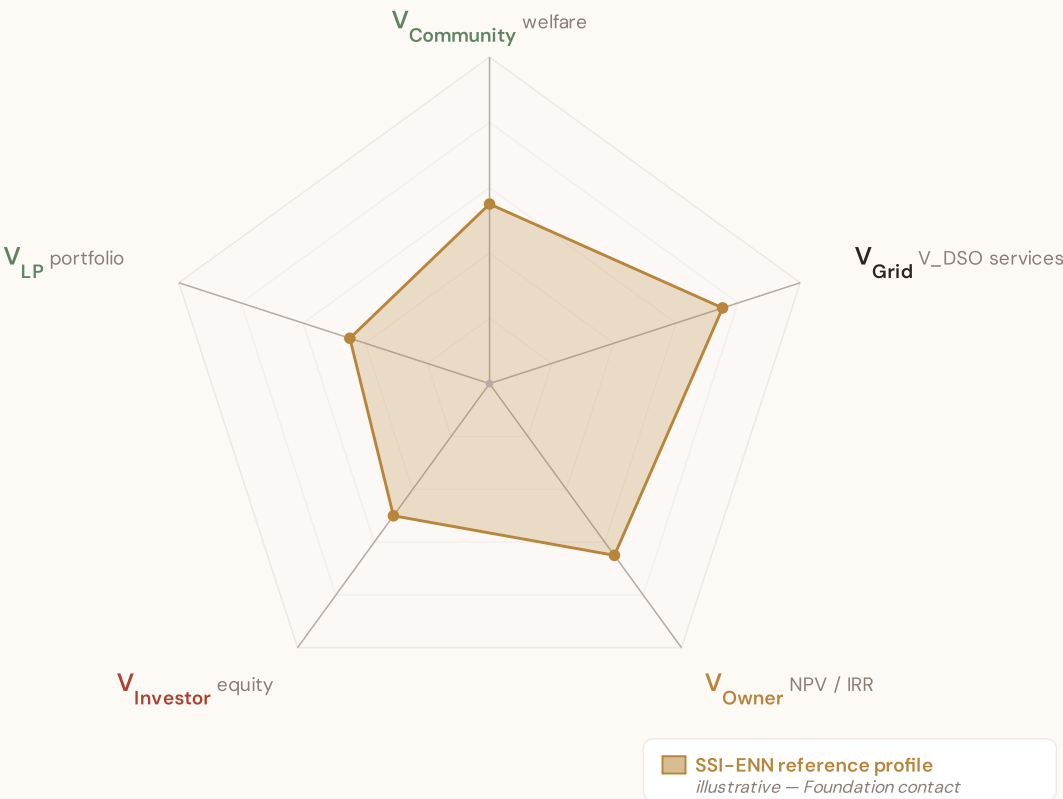


The Bogotá D.C. — capital district + Guavio supply corridor (Edition 01 inaugural) ranks **#0 of 28 departamento regions** by mean  $R_{\text{median}}$ , placing it among the upper tier of national risk. The three most stressed are bogota, santander, antioquia, while the three most resilient are guaviare, choco, arauca. 18 of 28 score above the fleet average of 0.467. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DSO or national statistics — reveals the true spatial distribution of grid stress. It is precisely this substation-level granularity that positions the SSI as a tool for investment prioritisation: not treating entire regions as monoliths, but identifying the specific corridors within each that demand attention.

**From grid intelligence to BESS valuation platform.** The SSI Index answers "How healthy is this substation?" The SSI Enhanced Neural Network (SSI-ENN) extends this into: "What is a BESS worth at this substation — to the DNO and to the community?" Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.

SSI-ENN value-articulation radar — 5-lens stakeholder taxonomy

The SSI-ENN's commercial-tier value articulation answers "what is a BESS worth at this substation — and to whom?" through five stakeholder lenses (architecture memo §5 Decision #12): **V<sub>Community</sub>** (welfare avoided outages, energy-poverty relief, employment) · **V<sub>Grid</sub>** (V\_DSO — avoided reinforcement, reliability, voltage, congestion, frequency, black-start, hosting capacity) · **V<sub>Owner</sub>** (NPV / IRR / payback per asset class) · **V<sub>Investor</sub>** (risk-adjusted equity return at the financing tier) · **V<sub>LP</sub>** (limited-partner cash-on-cash + DPI / TVPI portfolio metrics). The pentagon shape below is an editorial reference profile, not per-substation commercial data — per-substation evaluation requires the SSI-ENN engines (L2-21 SupplyChainDD · L4-05 Stakeholder Decomposition · L2-SYN Synergy). Foundation contact: [ssi\\_index@ikenga.eu](mailto:ssi_index@ikenga.eu).



*Editorial reference shape — V<sub>Grid</sub> dominates (V\_DSO captures avoided reinforcement + reliability + voltage support + congestion relief + frequency + black-start + hosting capacity, the bulk of measurable BESS value at the substation locus per architecture memo §5). V<sub>Community</sub> + V<sub>Owner</sub> are co-secondary. V<sub>Investor</sub> + V<sub>LP</sub> reflect the financing-tier waterfall above V<sub>Owner</sub> NPV. Per-substation commercial decomposition lives in the SSI-ENN — mail to [ssi\\_index@ikenga.eu](mailto:ssi_index@ikenga.eu) for analytics.*

Layer 1 — Data

SSI v4.0 component taxonomy, normalisation, MC engine

Layer 2 — Valuation

DSO/TSO + Community value stacks, TCO, real options

Layer 3 — Neural Net

GBT+DNN ensemble, transfer learning, SHAP

Layer 4 — Coverage

Country prioritisation, DCI, registry, calibration

Layer 5 — Platform

Products, API, revenue model, unit economics



# Total Cost of Ownership Model

Stochastic cost modelling — because BESS economics are not deterministic

CAPEX follows a learning-curve distribution calibrated to Bloomberg NEF projections. OPEX is modelled as a mean-reverting process with seasonal components reflecting temperature-dependent degradation. Battery degradation follows a non-linear capacity fade model: each cycle reduces usable capacity, with rate depending on depth-of-discharge, temperature, and C-rate. The SSI data feeds this: substations with high II (temperature trajectory) face faster degradation. The Monte Carlo engine propagates cost uncertainty alongside revenue uncertainty.

## KEY CONCEPTS

- CAPEX learning curve calibrated to BNEF
- Non-linear battery degradation:  $f(\text{DoD}, \text{temperature}, \text{C-rate})$
- SSI II (climate trajectory) drives degradation forecast
- Full NPV distribution, not point estimate

## LIVE SSI DATA CONNECTION

Across the 744-substation fleet, the average consequence multiplier (R3) is 0.528, indicating significant socio-economic exposure. 309 substations (41.5%) are classified High or Critical — these are the highest-priority candidates for BESS deployment.

**Coming next month:** **LAYER 1** Gaussian Copula Monte Carlo Engine — *How 10,000 simulations per substation capture the correlation structure across grid risk factors*

## SECTION F.5 — V4.2 ANTI-MALADAPTATION GOVERNANCE GATE

# Radar 2 — W1–W10 Substation Community Value

**Wall 1 — Public W1–W10 framework.** The 10-axis Radar 2 expresses per-substation community-value baseline pre-intervention: *W1 Reliability · W2 Carbon · W3 Local Economic · W4 Climate Resilience · W5 Digital Inclusion · W6 Sovereignty · W7 Local Value Retention · W8 Communications · W9 Supply Security · W10 Compute Multiplier*. The discipline floor that the anti-maladaptation governance gate protects: *"no W axis worse off after intervention."*

## 5/1/1/3 4-tier audit-state classification (mesh resolution per Convention #6 §5.6)

### Published (5) — solid fill

W1 · W2 · W3 · W4 · W8

*Peer-reviewed methodology operational at fleet scale.*

### Published-baseline (1) ▲ — solid fill +

chevron corner marker

W9 — Supply Security

*Peer-reviewed academic basis*

*(Albert/Holme topology-only baseline) with acknowledged scope limit (Buldyrev/IEEE*

*493 — N-1 contingency dynamics not captured at baseline).*

### Baseline-only (1) — striped fill

W5 — Digital Inclusion

*Foundation methodology under development; baseline value defensible at v4.2.*

### Commercial (3) — dashed-faded

W6 · W7 · W10

*Commercial deliverables — please contact the Foundation.*

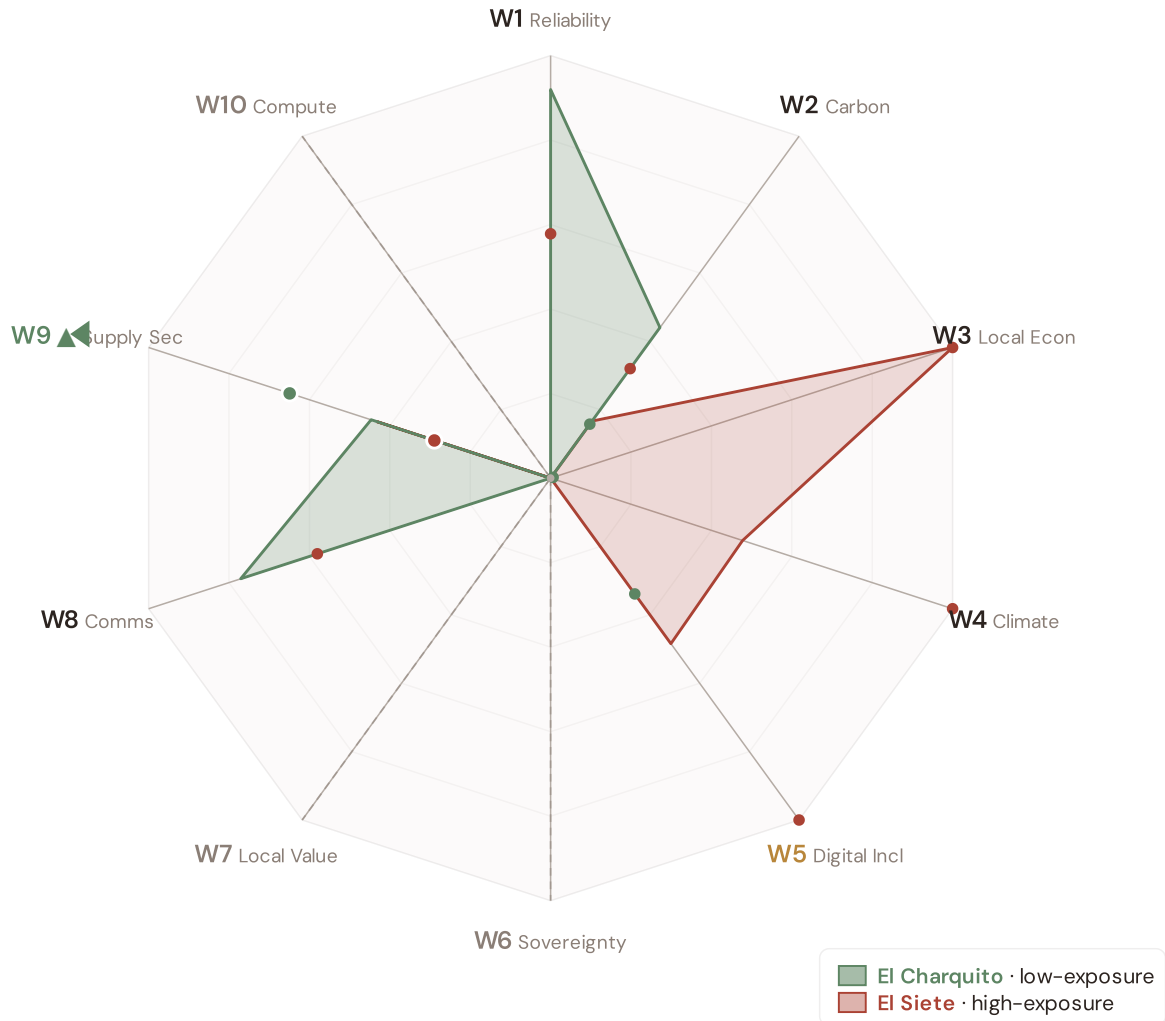
Per Convention #6 §5.6.7 academic-defensibility: the 4-tier classification aligns with IPCC AR6 + JRC COIN + Reckien (2021) continuum literature. The Published-baseline tier (W9) is the principled mesh resolution that honors both peer-review status + acknowledged scope limit without forcing a false binary.

## v4.2 methodology disclosure — per-country normalisation

Across Colombia, the substation fleet's component scores (C / V / I / E / S / T) and the Re composite have been normalised to [0, 1] via the canonical **P5/P95 percentile** method, using *per-country empirical bounds*. This is the same procedure Italy's v4.0.2 deployment applied at the scoring stage (SSI\_v4\_0 Italy/.../scoring-it/normalise.py); for v4.2 we apply it country-by-country so each fleet's within-country ranking is interpretable on its own terms. Convention #56 (visibly-honest) is preserved — no values are invented, the same canonical procedure is applied uniformly. Italy's pilot fleet keeps its native spread (0.17 – 0.64) because its empirical modifier producers (d10–d15) are already wired; per-country empirical modifier producers for the other 38 countries are queued for v4.5 and will refine the absolute risk ranking. Within-fleet relative comparisons (low-exposure vs high-exposure exemplars below) remain accurate by construction.

## Radar 2 visualization — anti-maladaptation governance gate

Two worked examples overlaid: **Bacata** (low-exposure reference,  $Re\_norm$  0.000) and **Rubiales** (high-exposure reference,  $Re\_norm$  1.000). Axis labels are colour-coded per the 5/1/1/3 mesh resolution; the ▲ chevron marks W9 as Published-baseline. Hover any axis label for the methodology tooltip.



*El Charquito polygon (sage): the low-exposure reference for this country — compact footprint reflects the fleet floor on the W-axis baselines. El Siete polygon (terracotta): the high-exposure reference — larger footprint where this country's fleet's W-axis values reach the upper percentile. Commercial-tier axes (W6 data sovereignty, W7 local value retention, W10 compute multiplier) display at zero per Convention #6 §5.6 mesh design — no public-data computation. W2 carbon uses EU-average MEF fallback (0.36) per v4.2 calibration.*

## W1–W10 axis enumeration with audit–state classification

5/1/1/3 4-TIER · V4.2 MESH RESOLUTION · LIVE VALUES VIA  
COMPUTE\_W\_BASELINES.PY

| AXIS | NAME                  | AUDIT STATE          | METHODOLOGY BASIS                                                                                                                                                                              |
|------|-----------------------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| W1   | Reliability           | Published            | $C \times E2_{\text{local}} \times \text{population} (SAIDI++-POV)$                                                                                                                            |
| W2   | Carbon                | Published            | $T1 \times \text{zonal MEF} \times (V \cdot \text{deg} / 100) (+ EEA)$                                                                                                                         |
| W3   | Local Economic        | Published            | $\text{norm}(R10_{\text{just}}) - \text{EU-SILC} + \text{OIPE energy-poverty}$                                                                                                                 |
| W4   | Climate Resilience    | Published            | $(R6d - 1) + (R6e - 1) + (R6c - 1) + \frac{1}{2}(R9 - 1) - v4.2$<br>resilience modifier family                                                                                                 |
| W5   | Digital Inclusion     | Baseline-only        | $\text{norm}(R8_{\text{adapt}}) - \text{bulk SM}++\text{DESI}$                                                                                                                                 |
| W6   | Sovereignty           | Commercial           | Foundation contact ( <a href="mailto:ssi_index@ikenga.eu">ssi_index@ikenga.eu</a> )                                                                                                            |
| W7   | Local Value Retention | Commercial           | Foundation contact ( <a href="mailto:ssi_index@ikenga.eu">ssi_index@ikenga.eu</a> )                                                                                                            |
| W8   | Communications        | Published            | $C \times p_{\text{crit\_10yr}} \times \text{population (Markov 10-yr critical probability)}$                                                                                                  |
| W9 ▲ | Supply Security       | Published–baseline ▲ | $0.60 \times BC_{\text{pct}} + 0.30 \times \text{bridge} + 0.10 \times (1 - \text{deg} / 20) -$<br>OSM power-graph betweenness centrality + bridge<br>identification + degree-based redundancy |
| W10  | Compute Multiplier    | Commercial           | Foundation contact ( <a href="mailto:ssi_index@ikenga.eu">ssi_index@ikenga.eu</a> )                                                                                                            |

### ▲ W9 Published–baseline — mandatory transparency tooltip

W9 — Supply Security (Published–baseline). Public-tier methodology: betweenness centrality + bridge identification + degree-based redundancy, computed deterministically from OpenStreetMap power-graph data. Peer-reviewed academic basis: Albert et al. (2000) *Nature*; Holme et al. (2002); broad power-grid centrality literature. Acknowledged scope limit (per Buldyrev et al. (2010) *Nature* + IEEE 493): topology-only baseline does not capture N–1 contingency dynamics or interdependent-network cascades. Richer methodology — N–1 contingency simulation + flexibility-services valuation — lives in SSI-ENN L2–22 / L2–23 commercial deliverables. Mail to [ssi\\_index@ikenga.eu](mailto:ssi_index@ikenga.eu) for extended analysis.

**Worked examples.** The W1–W10 per-substation baseline values for two anchor cases — El Charquito (low-exposure reference) and El Siete (high-exposure reference) — are computed live by `_audit-snapshot/scripts/d3_radar2_per_country.py` against the country's `ssi-data.json`. Per Convention #6 §2.3.6 Path B refactor (11 Jun 2026), this country's exemplars are now algorithmically selected (min/max `Re_norm`) rather than named picks. See the FC v2 reference for W-axis methodology + commercial-tier rationale.

**Anti-maladaptation discipline.** The 10-axis Radar 2 baseline is the per-substation floor that any post-intervention scenario must protect. No W axis may degrade after a BESS deployment, line upgrade, or operational change. The discipline is enforced per axis (not in aggregate), and per substation (not in fleet average). This is the v4.2 governance gate that ensures grid-modernisation projects deliver net community value without hidden trade-offs.

## SECTION G

# Looking Ahead

### NEXT MONTH — EDITION 04

#### Deep-Dive: Atlántico

Deep-dive into Atlántico's grid risk profile — substation map, component analysis, departamento regions breakdown.

### NEXT DATA REFRESH

#### Q4 2026 — o Quarterly Sources

**none** are due for refresh in Q4. Impact: DER registry (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **DANE — Departamento Administrativo Nacional de Estadística** (10 variables → R3 population, GDP per capita, poverty incidence (MPI), demographic structure).

### FLEET SPOTLIGHT

#### 39 Bridge Substations — Single Points of Failure

**39 substations** are topological bridges — their failure disconnects part of the network. Of these, **35** are classified High or Critical. These are the fleet's single points of failure: high graph centrality (R4) combined with elevated risk makes them priority candidates for both grid reinforcement and BESS deployment.

Edition 04 will be published on the second Thursday of September 2026. Deep-dive region: **Atlántico**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to [ssi\\_index@ikenga.eu](mailto:ssi_index@ikenga.eu).

**Feedback welcome.** The SSI Index is designed to evolve through stakeholder dialogue. If you represent an Colombian utility, , municipal energy company, or academic institution and would like to contribute data or methodology feedback, please contact us at [ssi\\_index@ikenga.eu](mailto:ssi_index@ikenga.eu). The SSI's credibility depends on open scrutiny.

## SSI Monthly Intelligence Report — Edition 006

Published 13 August 2026 · SSI Index **v4.2** (refreshed June 2026) · 381 substations · 2,278 grid lines · 6 components + **Re composite** (Radar 17th axis, bounds [0.920, 1.787]) · 20 metrics · 11 modifiers (5 baseline + 6 v4.2 resilience family)

10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196,608 evals/substation) · CMIP6 SSP2-4.5 · W1-W10 anti-maladaptation governance gate (5/1/1/3 4-tier mesh resolution per Convention #6 §5.6)

Foundation contact

The SSI Index is a non-profit foundation project, please mail to [ssi\\_index@ikenga.eu](mailto:ssi_index@ikenga.eu) for more details.